

Group Project 1

Group 26

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Contents

Introduction	2
Literature Review	2
Theory	3
Hypotheses	3
Data and Methods	3
Data	3
Dependent variable	3
Predictor variables	4
Exploratory Data Analysis	4
Predictor Variables	4
Data Cleaning	4
Methodology	7
Assumptions	7
Testing and Validation	7
Result Analysis	7
Plotting goodness of fit for each feature R^2 used in the 10-fold cross validated decision tree.	8
Testing Decision Tree Validity and Reliability	8
Correlation matrix	8
Skewness Checks	9
Detecting model for overfitting	10
Pre-pruning Exploration	10
Residual Analysis	11
Test-Retest Reliability	12
Limitations	13

Introduction

Quality of life refers to how well an individual perceives their life to be going. Over the past fifty years, it has become a central focus in social science research, driven by growing recognition that economic output alone poorly captures human wellbeing. Governments worldwide increasingly evaluate policy based on life outcomes rather than income alone, including Australia through its Measuring What Matters framework. While Australia consistently ranks highly in international wellbeing comparisons, meaningful inequality in quality of life persists within the population. This report uses data from the 2018 Australian World Values Survey to explore this variation. Five survey items measuring happiness, health, financial satisfaction, life satisfaction, and freedom of choice are combined into a composite measure using factor analysis. Key predictors examined include material deprivation, subjective social class, and neighbourhood trust. The central hypothesis is that lower deprivation, higher social class, and stronger community trust each predict significantly better quality of life.

Literature Review

To understand the factors that affect the quality of life of a person, it is important to refer core literatures that help build a concept related to the various topics and factors that relate to a person's quality of life. Early social science treated quality of life as roughly equivalent to income per capita. This view was challenged by Easterlin (1974), whose paradox demonstrated that average happiness rises only modestly as national income grows. Subsequent scholars consolidated the field: Diener et al. (1999) decomposed subjective wellbeing into positive affect, negative affect, and life satisfaction, while Ryff (1989) emphasised structural dimensions such as autonomy and personal growth. The WHO's WHOQOL project further established quality of life as a multidimensional latent construct requiring indirect measurement across physical, psychological, social, and environmental domains (WHOQOL Group, 1995; Veenhoven, 1996).

Some researchers argued that access to services and material deprivation are some factors affecting quality of life. This idea was supported by Townsend (1979), who argued that poverty is not about falling below an absolute income threshold, but about exclusion from the customary activities and comforts of one's society. Mack and Lansley (1985) and Nolan and Whelan (1996) refined this approach, showing that direct deprivation measures identify more disadvantaged populations than income measures alone. Three specific deprivations are examined here. Food insecurity is consistently associated with poorer mental health and reduced life satisfaction even after controlling for income (Gundersen & Ziliak, 2015), and remains non-negligible in Australia (Foodbank Australia, 2022). Going without needed medical treatment captures hardship even within universal healthcare systems, where out-of-pocket costs for specialists and medications lead many to forgo care — affecting around a fifth of Australian households despite Medicare (AIHW, 2020). Going without a cash income reflects acute liquidity crises and independently predicts chronic stress and reduced life satisfaction (Gallo et al., 2006). Crucially, these deprivation measures are conceptually distinct from income, capturing dimensions of hardship that income bands alone cannot.

There are also cultural and personal outlook on society and social standings that affect a person's evaluation of their quality of life. The Whitehall studies showed that health and wellbeing improve monotonically along occupational hierarchies, even where material hardship is absent — a pattern Marmot (2004) termed the status syndrome. Deaton (2008) and Kahneman and Deaton (2010) similarly found that life evaluation continues rising well into upper income levels. Subjective social class — where individuals locate themselves in the hierarchy — predicts wellbeing above and beyond objective income, reflecting both its role as a summary of economic and social resources and its independent psychosocial effects (Singh-Manoux et al., 2005; Adler et al., 2000). Australian research confirms that distinct class clusters exist domestically and differ systematically in their social and economic resources (Sheppard & Biddle, 2017). Putnam (2000) and

Helliwell and Putnam (2004) demonstrated that social trust and civic participation predict life satisfaction as strongly as income. Neighbourhood trust in particular shapes perceived safety, social support, and mental health (Kawachi et al., 1999), with Australian evidence linking lower neighbourhood trust to reduced quality of life (Markus, 2020).

Theory

Quality of life in wealth societies like Australia is not determined by national prosperity alone, but by three layered and partially independent dimensions of individual experience: what people have access to related to material security and social services, where they view their social standings in society and the trust they have in their local community and society. Deprivation in any one of these dimensions independently undermines the wellbeing, and cumulative disadvantage across the dimensions compound the effect.

Hypotheses

The five predictors examined in this report are drawn directly from the three traditions reviewed above. Our hypotheses are as follows.

- H1 (Material deprivation): Respondents who rarely or never experience going without food (V188), medical treatment (V190), or cash income (V191) will report higher quality of life than those who frequently do, with each deprivation carrying an independent effect as each reflects a conceptually distinct hardship.
- H2 (Socioeconomic position): Respondents who place themselves higher in the Australian class structure (V236) will report higher quality of life, reflecting the psychosocial effects of perceived status and social position rather than acute material hardship.
- H3 (Social capital): Respondents who express greater trust in their neighbours (V103) will report higher quality of life, consistent with Putnam's social capital framework and evidence linking neighbourhood trust to wellbeing.

Data and Methods

Data

This analysis draws on the 2018 Australian World Values Survey (WVS Wave 7), which surveyed 1,813 Australians via online and hard-copy questionnaires. The survey covers a broad array of social attitudes, values, and life circumstances. Data were obtained from the Australian Data Archive.

Dependent variable

Quality of life is operationalised as a latent factor score derived from a single-factor factor analysis of five items: self-reported happiness (V10), health (V11), freedom of choice (V55), life satisfaction (V23), and financial satisfaction (V59). Responses are recoded so that higher scores indicate better outcomes, and missing values (coded -2) are set to missing before analysis.

Descriptive Statistics of QoL

minimum	q1	median	mean	q3	maximum	na
-3.926787	-0.4393202	0.1744014	-0.001894926	0.6510221	1.506232	55

Predictor variables

Each predictor is cleaned by replacing the —2 (no answer) code with missing values. Two variables - social class (V236) and trust in neighbours (V103) are reverse-coded so that higher values uniformly represent better outcomes, consistent with the three deprivation indicators.

- **Food security** (`food_secure`, V188): How often the respondent has gone without enough food to eat (1 = Often, 4 = Never). Higher values = greater food security.
- **Medical access** (`medical_access`, V190): How often the respondent has gone without medicine or medical treatment they needed (1 = Often, 4 = Never). Higher values = better access.
- **Income security** (`income_security`, V191): How often the respondent has gone without a cash income (1 = Often, 4 = Never). Higher values = greater income stability.
- **Social class** (`social_class`, V236): Self-identified class position. Originally coded 1 = Upper class to 5 = Lower class; reverse-coded so 5 = Upper class, 1 = Lower class.
- **Social trust** (`social_trust`, V103): Trust in people in the neighbourhood. Originally coded 1 = Trust completely to 4 = Do not trust at all; reverse-coded so 4 = Trust completely.

Exploratory Data Analysis

The descriptive statistics of the quality of life score show that the average person is unhappy (Score < 0), this is supported by the histogram and at Figure 1 showing the distribution has a negative skew present in the dataset. Although the dataset is large enough for the Central Limits theorem to be applied ($n = 1813$), it is difficult to capture a linear relationship of the predictor variables and quality of life, and hence we preferred to model quality of life by capturing non-linear relationships.

The dataset contains around 1.5% missing entries as can be seen in Figure 2. To reduce the model complexity of our decision tree and hence reduce the chance of overfitting in the decision tree, we removed NULL values during the data cleaning process.

From Figure 4, we see that aside from neighbourhood safety, the class distributions are imbalanced, which could harm the validity of a decision tree-based approach.

Predictor Variables

As all attributes in the dataset are categorical in nature, we investigated the relationship between the selected attributes with quality of lifestyle score using boxplots. From the boxplot in Figure 3, we see that the Quality of life improves as circumstances improve (plentiful access to food, feeling safe in your environment, cash in hand available). While first four boxplots (QoL vs Food - Gone without Cash Income) show a trend down, it's important to note that these were measured inversely. I.e. Larger = Better.

Data Cleaning

The cleaning of the data was done using the following steps: - Since the Quality of Life feature was the dependent variable for our research, we removed all the rows where the dependent variable was missing.

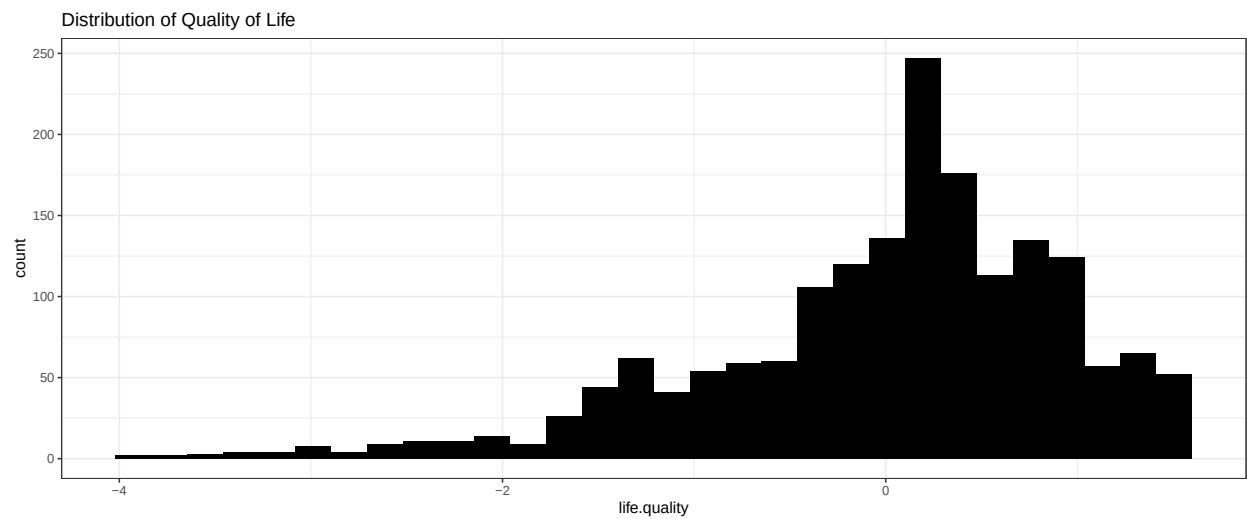


Figure 1: Histogram showing the distribution of Life Quality, the distribution has a negative skew

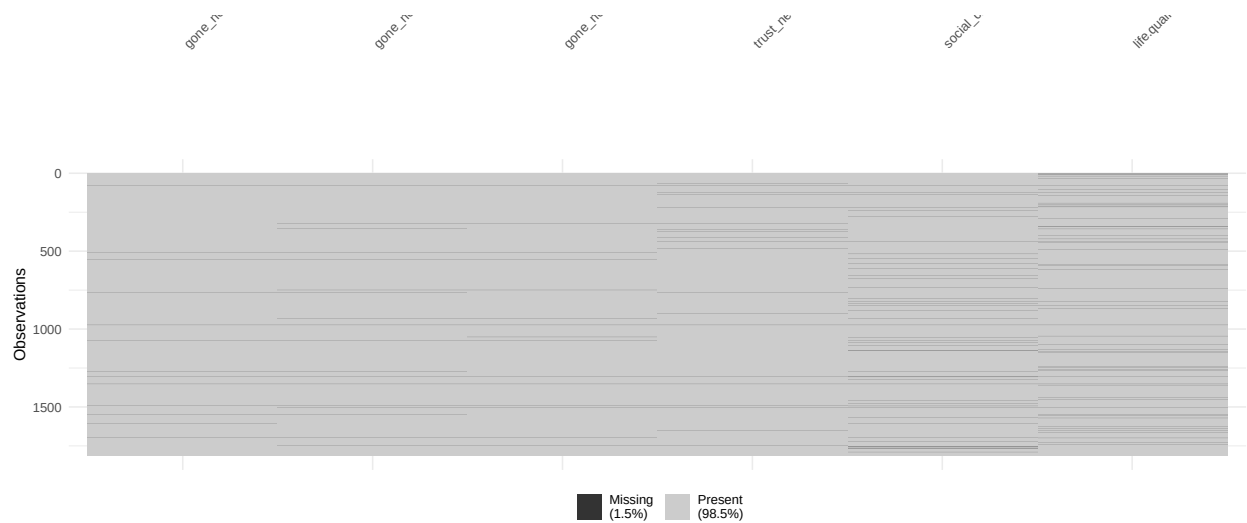


Figure 2: Visualisation of missing data in the dataset, there are only a small amount of missing entries present in the dataset.

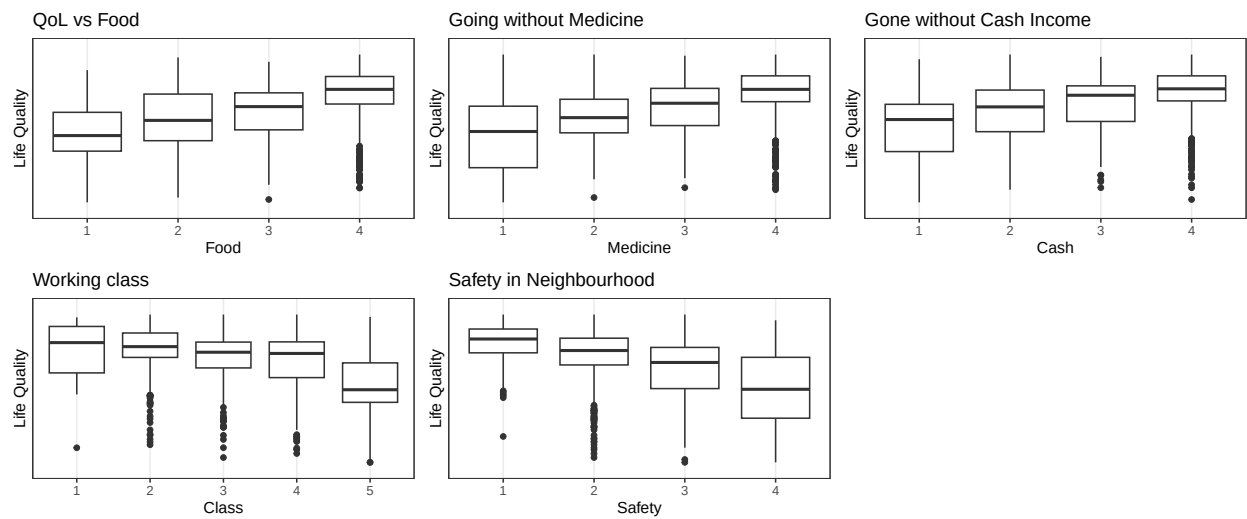


Figure 3: Boxplots of predictor variables against Quality of Life, scores of safety and class are inverted i.e. 1 is highest

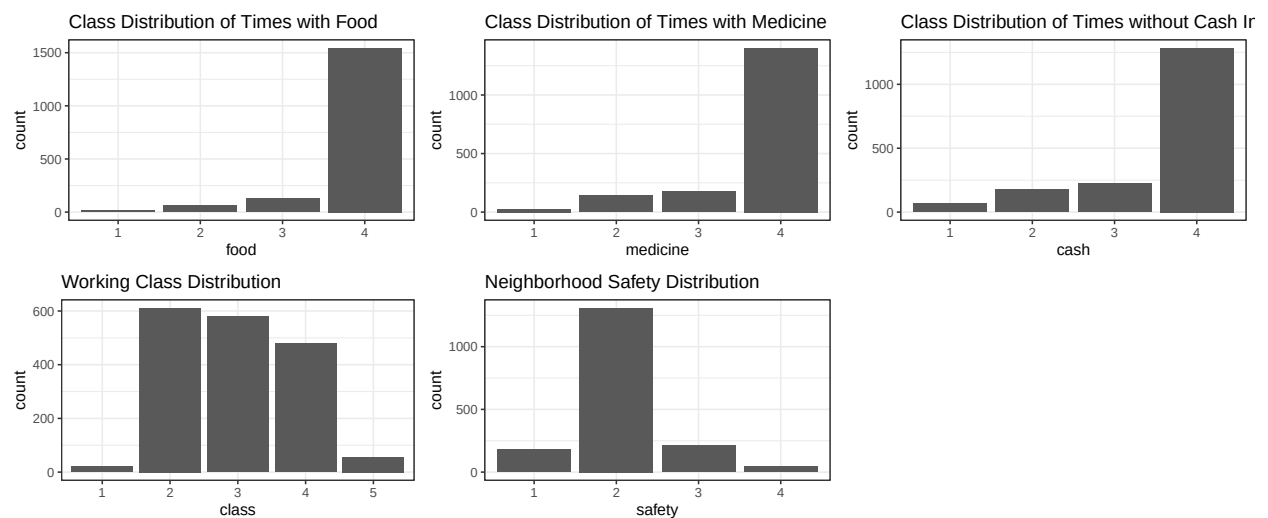


Figure 4: Class Distribution of variables, there is a high amount of class imbalance in the dataset.

- For the records with missing values in the other columns, the count was less than 20 - so we decided it would be logically easier to remove those records as well. - As all the features were categorical, it was difficult to detect outliers in the independent variables, so we decided to leave it as such since the imbalance in independent features would help us understand more about the demographic of the survey.

Methodology

Assumptions

Given the observed negative skew in the quality of life distribution, a decision tree model was selected as it makes no parametric assumptions about the distribution of the dependent variable. Unlike OLS regression, decision trees partition the feature space through recursive binary splits and are therefore robust to non-normality in the outcome variable, making them appropriate for the skewed latent factor scores observed in this sample.

However, the features in the data may have a non-linear relationships with the latent variable. This may be due to the low number of features selected which may cause compounding variables to influence the result.

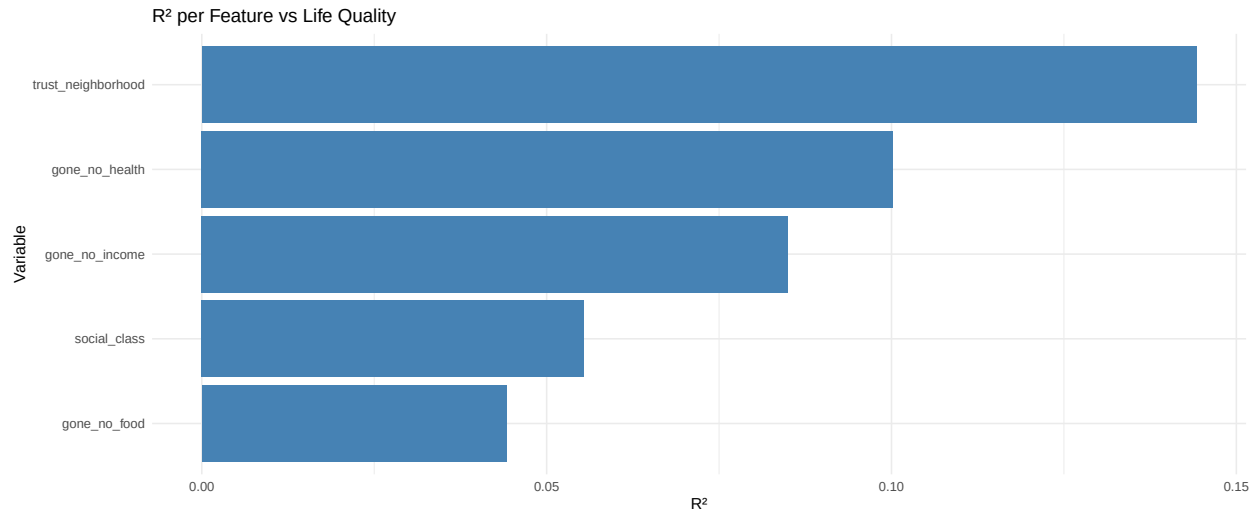
Testing and Validation

To test our hypotheses and validate our assumptions we will test 3 different decision trees to come to a concrete conclusion regarding the applicability of decision trees to this research question and how much importance each feature (gone with out income, gone without food, gone without medicine, trusting neighbors and social class) affects the latent variable Quality of life.

The three decision trees validation that we will be testing are: - Comparing the pruned and unpruned decision trees to see if the features that are selected are reliable or not. - Using K-folds cross validation with decision trees to see if there are any relation between the features and the latent variable. - Using an ensemble of decision trees (bagging) to see how different combinations of the same feature set may lead to different results.

Result Analysis

Plotting goodness of fit for each feature R^2 used in the 10-fold cross validated decision tree.

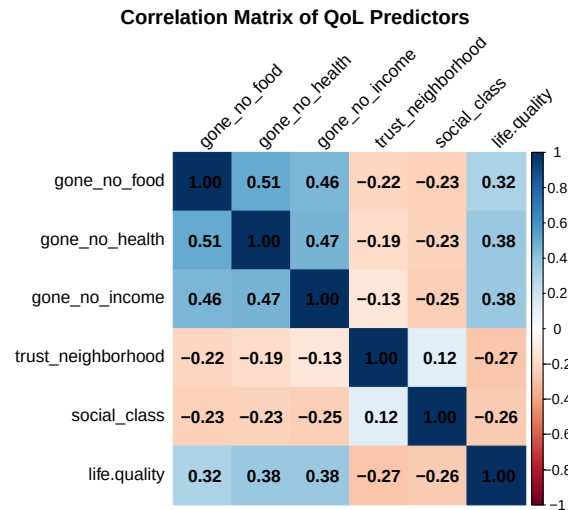


The low R^2 observed may reflect both the non-linear nature of relationships between predictors and quality of life, and the influence of omitted variables not captured by the five predictors selected, warranting cautious interpretation of the results.

Testing Decision Tree Validity and Reliability

Correlation matrix

```
##               gone_no_food gone_no_health gone_no_income
## gone_no_food         1.0000000      0.5062064      0.4554664
## gone_no_health       0.5062064      1.0000000      0.4688423
## gone_no_income       0.4554664      0.4688423      1.0000000
## trust_neighborhood  -0.2173007     -0.1879949     -0.1325449
## social_class        -0.2343073     -0.2252231     -0.2453556
## life.quality         0.3188351      0.3757251      0.3769237
##
##               trust_neighborhood social_class life.quality
## gone_no_food        -0.2173007    -0.2343073    0.3188351
## gone_no_health      -0.1879949    -0.2252231    0.3757251
## gone_no_income      -0.1325449    -0.2453556    0.3769237
## trust_neighborhood   1.0000000     0.1213683   -0.2721205
## social_class          0.1213683     1.0000000   -0.2554861
## life.quality         -0.2721205    -0.2554861     1.0000000
```

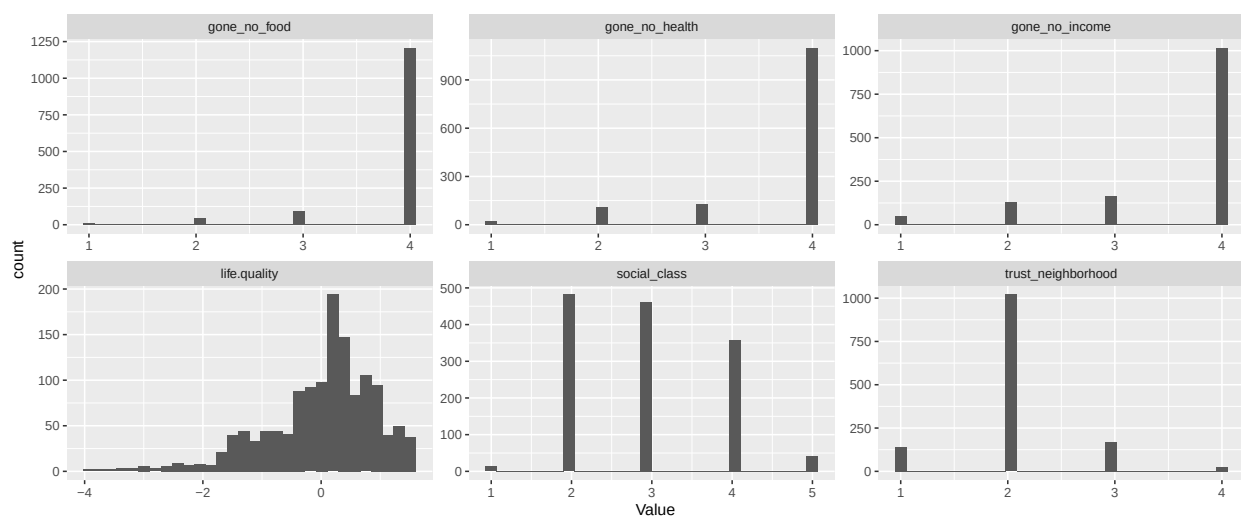



Discussion

- Positive correlation between health, income and food although the relationship itself is quite weak.
- All three variables have mildly positive relationships with each other, indicating that these factors affect each other.
- Very weak positive relationship between trust in neighbourhoods and social class.

Skewness Checks

```
##                skewness
## gone_no_food      -3.447
## gone_no_health    -2.234
## gone_no_income    -1.835
## life.quality       -1.034
## trust_neighborhood  0.747
## social_class       0.281
```

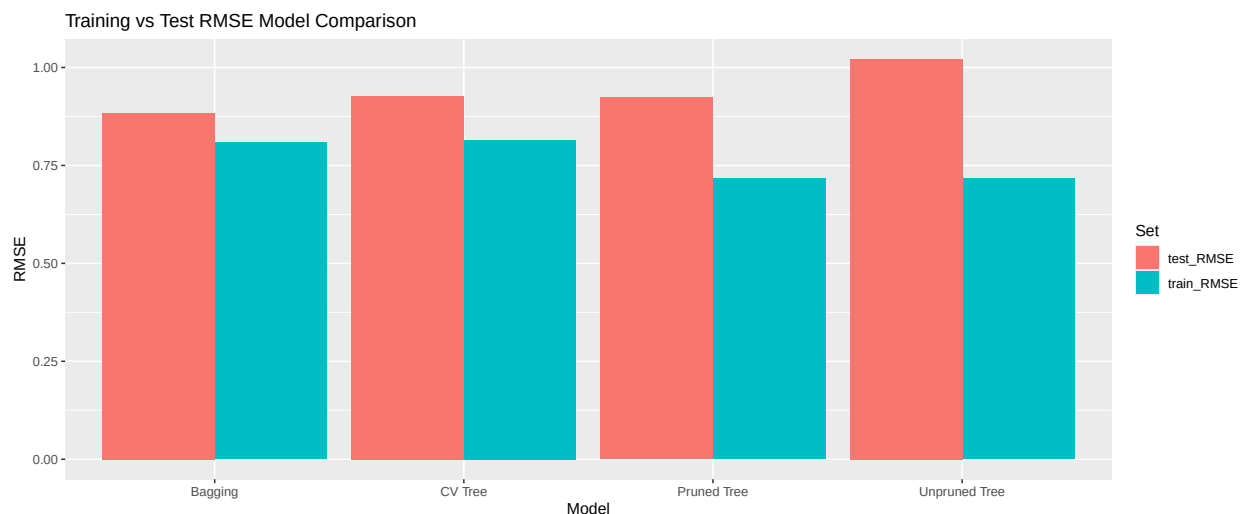


Discussion

- For variables `gone_no_food`, `gone_no_health`, `gone_no_income`, majority of survey responses sit at highest possible score. This also confirms left-skew distribution for `life.quality`.
- Very few respondents indicated poor quality of life factors (not enough food, bad health and low income).
- Indicates that ceiling effect might be occurring, a measurement limitation that is occurring in the survey due to a low upper limit.
- Graph distribution of `life.quality` values show a left skew, these match skewness values shown above.
- `social_class` more balanced, values concentrated on response values 2-4

Detecting model for overfitting

##	Model	train_RMSE	test_RMSE	RMSE_gap
## 4	Bagging	0.8083	0.8840	0.0757
## 3	CV Tree	0.8153	0.9269	0.1116
## 2	Pruned Tree	0.7171	0.9231	0.2060
## 1	Unpruned Tree	0.7171	1.0222	0.3051



Discussion

- Tree trained with cross-validation saw the smallest gap in RMSE, indicating this model strongly generalises on unseen data.
- Unpruned Tree model had the largest gap between train and test RMSE, indicating this model is more prone to overfitting on unseen data.
- Both pruned and unpruned tree saw the same RMSE on training data, however the pruned tree was less error-prone on the test data, showing the effectiveness of pruning the decision tree.

Pre-pruning Exploration

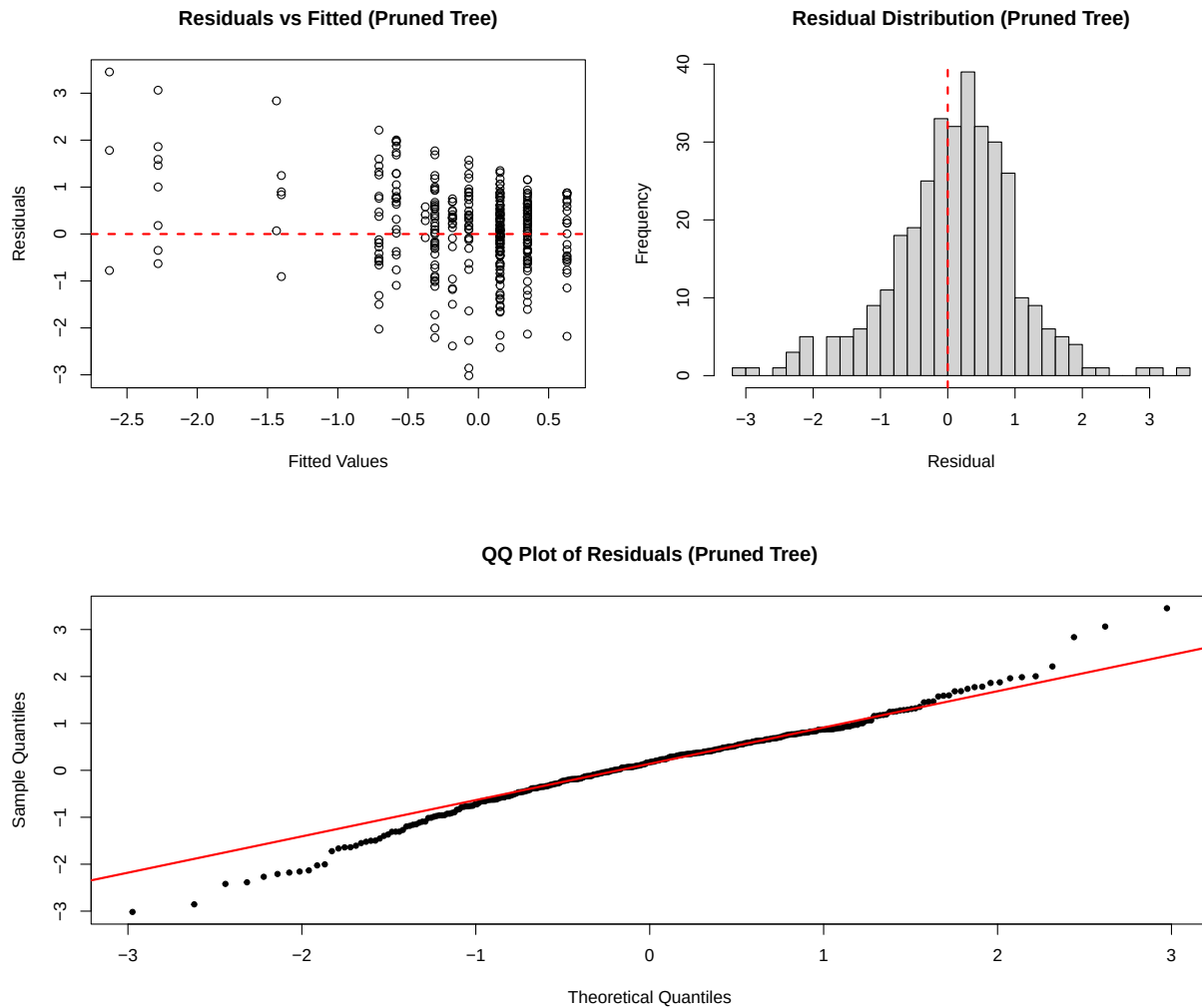
##	MaxDepth	MinSplit	MinBucket	RMSE	R2	N_leaves
## 4	15	30	15	0.9218	0.0531	7

## 2	5	10	5	0.9272	0.0421	8
## 3	10	20	10	0.9272	0.0421	8
## 1	3	10	5	0.9315	0.0333	7

Discussion

- Pre-pruning reveals that the decision tree naturally stops growth, produces 7-8 leaves regardless of what is specified as the maximum depth of the tree.
- Consistently low R^2 value, ranging from ~ 0.03 to ~ 0.05 . Ties back into the point raised about ceiling effects
- All variants reveal similar RMSE values

Residual Analysis



Residual Summary:

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	-3.01721	-0.38205	0.17301	0.09829	0.66114	3.45295

```
##
## Mean Residual: 0.0983

## SD of Residuals: 0.9192
```

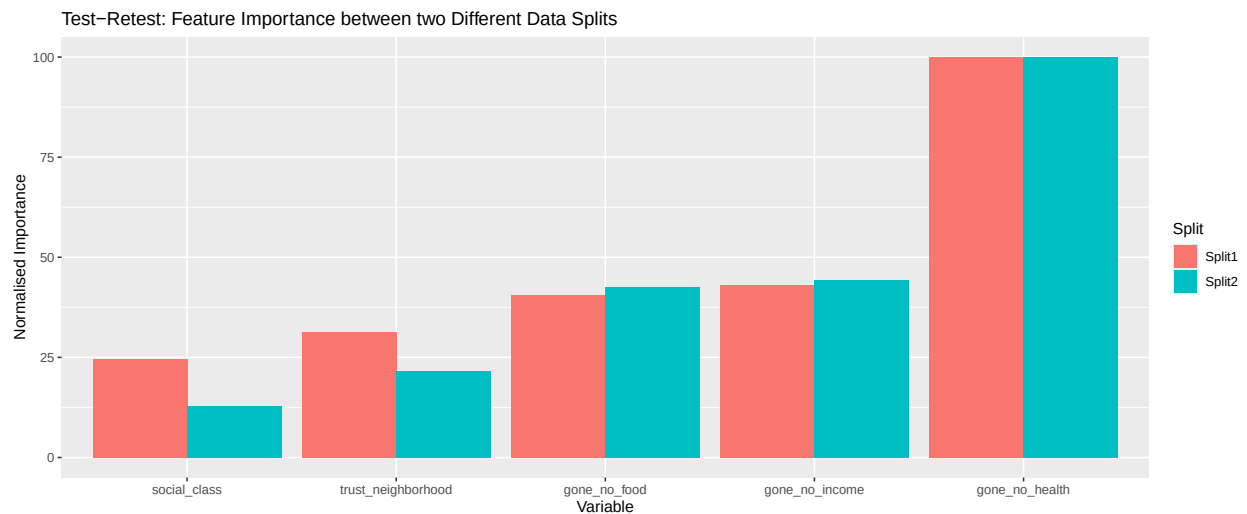
Discussion

- Residual vs fitted plot displays no systemic bias, points follow shape of vertical columns but this is expected from survey data and decision tree nature (fixed leaf nodes).
- Residual distribution shows a very close bell-shape, mostly centered near zero.
- QQ Plot confirms near-normality of data, points follow diagonal very closely in the middle and minimal deviation at both tails of the graph
- Mean residual of 0.098 shows slight tendency to under-predict QoL.
- Plots show that model is overall valid, especially with how distribution of residuals how the QQ Plot is shaped.

Test-Retest Reliability

```
##
## Variable Split1 Split2
## gone_no_health gone_no_health 100.00 100.00
## gone_no_income gone_no_income 43.19 44.26
## gone_no_food gone_no_food 40.53 42.59
## trust_neighborhood trust_neighborhood 31.29 21.53
## social_class social_class 24.63 12.94

## Feature Importance Rank Correlation (Spearman): 1.000
```



Discussion

- Strong consistency in rankings between the two splits
- gone_no_health the most dominant predictor (scoring 100 in both splits)

- `gone_no_income` and `gone_no_food` demonstrated strong stability in both splits, minimal score deviation from first split to second split (~1 point difference for `gone_no_income` and ~2 point difference for `gone_no_food`)
- `trust_neighbourhood` and `social_class` showed greater variation in feature importance score, still retained same rank in both splits.

Limitations

The non-linear relationships between predictors and quality of life reduce the suitability of tree-based models, with factor analysis or neural network approaches likely yielding stronger results. Additionally, examining only five of fifty-plus available variables introduces omitted variable bias, limiting the model's comprehensiveness and suggesting the need for further research.

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